

Global EV Outlook 2025

Expanding sales in diverse markets

International
Energy Agency



Abstract

The Global EV Outlook is an annual publication that reports on recent developments in electric mobility around the world. It is developed with the support of members of the Electric Vehicles Initiative (EVI).

The report draws on the latest data to assess trends in electric vehicle deployment, demand for their batteries and charging infrastructure. It considers recent policy developments and industry strategies shaping the outlook for electric vehicles in different markets. This edition features analysis of electric vehicle affordability, manufacturing and trade of electric cars and their batteries, and the total cost of ownership of electric heavy-duty trucks across various markets, and makes projections to 2030.

The report is complemented by updated versions of two online tools: the Global EV Data Explorer and the Global EV Policy Explorer, which allow users to further explore EV statistics and projections, and policy measures worldwide.

Executive summary

Electric car sales continue to break records globally, particularly in China and other emerging economies

Electric car sales exceeded 17 million globally in 2024, reaching a sales share of more than 20%. Just the additional 3.5 million electric cars sold in 2024 compared with the previous year is more than the total number of electric cars sold worldwide in 2020. China maintained its lead, with electric cars accounting for almost half of all car sales in 2024; the over 11 million electric cars sold in China last year were more than global sales just 2 years earlier. As a result of continued strong growth, 1 in 10 cars on Chinese roads is now electric. Europe saw sales stagnate in 2024 as subsidy schemes and other supportive policies waned, but the sales share of electric cars remained around 20% as stronger sales in some countries compensated for lower sales in others. In the United States, electric car sales grew by about 10% year-on-year, reaching more than 1 in 10 cars sold.

Emerging markets in Asia and Latin America are becoming new centres of growth, with electric car sales jumping by over 60% in 2024 to almost 600 000 – about the size of the European market 5 years earlier. In Southeast Asia, electric car sales grew by nearly 50% to represent 9% of all car sales in the region, with notably higher sales shares in Thailand and Viet Nam. In Brazil, the largest car market in Latin America, electric car sales more than doubled to 125 000 in 2024, reaching a sales share of over 6%. Sales in Africa also more than doubled, too, mostly thanks to growing sales in Egypt and Morocco, though electric cars still represent less than 1% of total car sales across the continent. Policy support and relatively affordable electric car imports from China played a central role in increasing sales in some emerging electric vehicle (EV) markets, accounting for 85% of electric car sales in both Brazil and Thailand, for example. Across all emerging economies outside of China, Chinese imports made up 75% of the increase in electric car sales in 2024.

Expected growth in electric car sales is led by China and Europe, alongside a surge in emerging economies

Electric car sales in 2025 are expected to exceed 20 million worldwide to represent more than one-quarter of cars sold worldwide. Sales were up 35% year-on-year in the first three months of 2025, with record first-quarter sales in all major markets. In China, the continuation of incentives for replacing older vehicles and falling electric car prices mean electric cars are projected to reach around

60% of total car sales in the country in 2025. Emissions standards in the European Union and the United Kingdom will require higher shares of zero-emission car sales in 2025. Building on more than 20% year-on-year sales growth observed in the first quarter, these policy pushes are expected to drive up electric car sales in Europe in 2025 to reach a sales share of 25%, despite flexibility given to automakers for meeting the 2025 EU emissions reduction target. While the 2025 outlook for electric car sales in the United States is uncertain based on today's policy direction, sales are currently expected to maintain the 10% growth observed in the first quarter; as consumers take advantage of existing tax credits in view of their potential repeal, electric car sales are projected to reach 11% of total car sales over the full year. In emerging economies other than China, sales are expected to continue growing strongly, increasing by 50% to reach 1 million in 2025.

Despite uncertainties in the outlook, the share of electric cars in overall car sales is set to exceed 40% in 2030 under today's policy settings. China is poised to continue leading in electric car sales to 2030, achieving a sales share of around 80% on the back of significant market momentum and competitively-priced EVs. In Europe, carbon dioxide (CO₂) targets support the achievement of a sales share of close to 60% by 2030, slightly below last year's projection. The sales share in the United States grows much more modestly than in our *Outlook* last year, reaching around 20% by 2030 based on today's policy direction – less than half the share projected for 2030 last year. In Southeast Asia, meanwhile, electric car sales are boosted by strong policy support and available domestic manufacturing capacity: by 2030, one in four cars sold in the region is poised to be electric. Two/three-wheelers – an important mode of road transport in the region – electrify faster: by 2030, almost 1 in 3 such vehicles sold are electric. Across all vehicle modes, the deployment of EVs replaces the use of more than 5 million barrels of oil per day globally in 2030, an important energy security consideration. Half of these savings are the result of EV adoption in China.

Uncertainty about the evolution of trade and industrial policy, downside risks to the economic outlook, and lower oil prices could affect EV uptake – but also car markets overall. Higher tariffs might increase the price of cars, including electric cars, and their components; lower GDP growth could dampen car sales; and lower oil prices affect the fuel cost savings from the use of electric cars. The way these factors will play out in practice is uncertain, but on aggregate they look to pose risks for overall car sales volumes more than for the share of EVs. In China, continued political support and competitive EV prices suggest that EV sales can withstand such headwinds. In Europe, where price differentials with conventional cars are larger, the combination of longer-term policy ambition and examples of policy responses from the pandemic suggest that sustaining EV sales is possible. Low oil prices can reduce fuel savings offered by battery electric cars, but even at global benchmark oil prices of USD 40 per barrel, there would be

significant fuel cost savings in all major EV markets when charging at home. In China, public fast charging costs around twice as much as home charging, but it would still offer savings for EV owners compared with driving a conventional car.

Global trade of electric cars is growing as manufacturers eye new markets

China continues to be the world's EV manufacturing hub and is responsible for more than 70% of global production. Car manufacturers headquartered in China predominantly cater to the domestic market, accounting for around 80% of domestic sales in 2024 and almost all of the 25% growth in global EV production. In the European Union, production stalled at 2.4 million electric cars in 2024. North America saw contrasting trends: US production declined whereas Mexico's output doubled, supported by comparatively low manufacturing costs. Roughly 70% of Mexico's output was from US-headquartered manufacturers. Production also increased by 15% in Asia Pacific countries other than China, reaching about 1 million electric cars, mostly from incumbent carmakers from Japan and Korea.

Global trade of electric cars increased 20% in 2024; imports now represent almost one-fifth of global electric car sales. At 40% (nearly 1.25 million electric cars), China accounted for the largest share of global exports in 2024. The European Union also remained a net exporter of electric cars: exports reached over 800 000, mostly destined for other European countries (such as the United Kingdom) and North America. EU imports were below 700 000, of which 60% came from China. The United States remained a net importer of electric cars; imports increased by nearly 40% in 2024, while exports fell by nearly 15%.

Chinese EV export markets are diversifying as Chinese automakers make headway in Brazil, Mexico and Southeast Asia. Several potential markets for Chinese EV exports have recently adopted or are considering the use of tariffs. This has prompted Chinese manufacturers to either frontload their exports before such tariffs come into force (as in Brazil) or to seek new markets for their available output. Today, overseas manufacturing capacity from Chinese manufacturers supplies about 5% of electric car sales in emerging markets and is set to grow further.

Competition and declining battery prices are improving affordability, though progress is uneven

As a global average, the price of battery electric cars fell in 2024, but the purchase price gap with conventional cars persisted in many markets. The average battery electric car price in Germany, for example, remained 20% higher than that of its conventional counterpart. In the United States, battery electric cars remained 30% more expensive, dampening future sales growth expectations. In

contrast, two-thirds of all electric cars sold in China in 2024 were priced lower than their conventional equivalents, without considering purchase incentives for EVs. This helped boost sales even as government incentives decreased.

Chinese electric car models are typically cheaper than the average EV in emerging markets, bolstering the competitive position of the Chinese industry. In Thailand, the average price of a battery electric car has now reached parity with an average conventional car, and the Chinese electric cars available are, on average, even cheaper. In Brazil, the price gap between battery electric cars and conventional cars shrank to 25% in 2024 from over 100% in 2023 as Chinese electric car imports grew to 85% of the country's EV sales. Similarly, the average price premium of battery electric cars in Mexico fell to 50% in 2024 from more than 100% in 2023 as Chinese imports reached two-thirds of sales.

Low critical mineral prices and increasing competition between battery manufacturers drove down battery pack prices in all markets in 2024, albeit with significant variations. In China, prices fell by about 30%, compared with 10-15% in Europe and the United States. The faster pace of cost reductions in China – enabled by strong competition, increasing manufacturing efficiency and supply chain integration, and access to a skilled workforce – is increasing the competitive advantage of Chinese battery manufacturers.

Improving charging coverage, capacity and integration is key to growing adoption of electric vehicles

Public charging stations have doubled in the past 2 years to keep up with growing EV sales. China and the European Union have maintained a steady pace of charger deployment compared with the number of EVs on the road. However, in both the United States and United Kingdom, which have higher rates of access to home chargers than China, public charger build-out has not kept pace with EV deployment, and the number of electric light-duty vehicles per public charging point increased in 2024.

Charging capacity is an important measure of the adequacy of public charging networks. The number of ultra-fast chargers, with power ratings of 150 kilowatts (kW) and above, grew by about 50% in 2024 and now account for nearly 10% of all public fast chargers. Public slow chargers in urban areas are a solution for EV owners without access to home charging, but fast chargers along highways help enable long-distance trips. In Europe, over three-quarters of all highways have a fast-charging station at least every 50 kilometres, compared with less than half of US highways. Globally, public charging capacity for light-duty EVs would need to grow by almost ninefold to 2030 to support EV sales implied by stated policies. Even so, EVs are set to account for only 2.5% of total global electricity demand in 2030.

Government efforts to promote interoperability, standardisation, smart charging and vehicle-to-grid integration can ease the transition to EVs for drivers and grid operators. Technologies such as smart chargers and vehicle-to-grid-capable EV models are becoming more available, but new market structures and legal frameworks will be needed for vehicles to maximise the potential benefits for the grid. China and the United Kingdom stand out for policy implementation and demonstration projects in this area. Battery technology innovations in recent years are also enabling safe high-power charging that could be as quick as refuelling a conventional car, but fully realising these advances hinges on deploying adequate infrastructure.

The value proposition for electric trucks is improving, even for long-haul operations

Electric truck sales grew by nearly 80% globally in 2024 to reach close to 2% of total truck sales. Spurred by a new scrappage scheme for conventional trucks, China's electric truck sales doubled to 75 000 vehicles, representing over 80% of global sales in 2024. In Europe and the United States, electric truck sales in 2024 remained similar to 2023 levels. The number of battery electric truck models available has grown from less than 70 in 2020 to more than 400, increasing the number of applications that can be met by electric trucks.

The total cost of ownership of a battery electric heavy-duty truck is already lower than for a diesel equivalent in China in certain cases. The purchase price of battery electric heavy-duty trucks remains two to three times higher than equivalent diesel trucks in major markets, which can discourage fleet operators, who often operate under tight margins. However, greater efficiency and lower energy costs – even including the cost of a high-power charger – make battery electric trucks more attractive the more they are used. By 2030, battery electric trucks in Europe and the United States are expected to reach parity on the total cost of ownership with diesel trucks for long-haul operations, as they already have in China, and are set to remain more cost-effective than hydrogen fuel cell trucks. However, the specific application and use profile of trucks are key factors for determining which powertrain technology is most suitable, meaning case-by-case analysis may be needed to evaluate the costs of various alternatives.

Mandated rest periods for truck drivers can play an important role in reducing the opportunity cost of charging en route. For long-haul trucking, the wait time for recharging an electric truck can pose difficulties for logistics operators. However, the 45-minute rest period mandated in the European Union can accommodate the addition of around 150 km of driving range for a heavy-duty truck using a 350 kW charger, and up to about 400 km if a megawatt charger is used. While the United States and China also have mandated rest periods, the design of the EU policy is currently most conducive to supporting EV adoption.

1. Trends in electric car markets

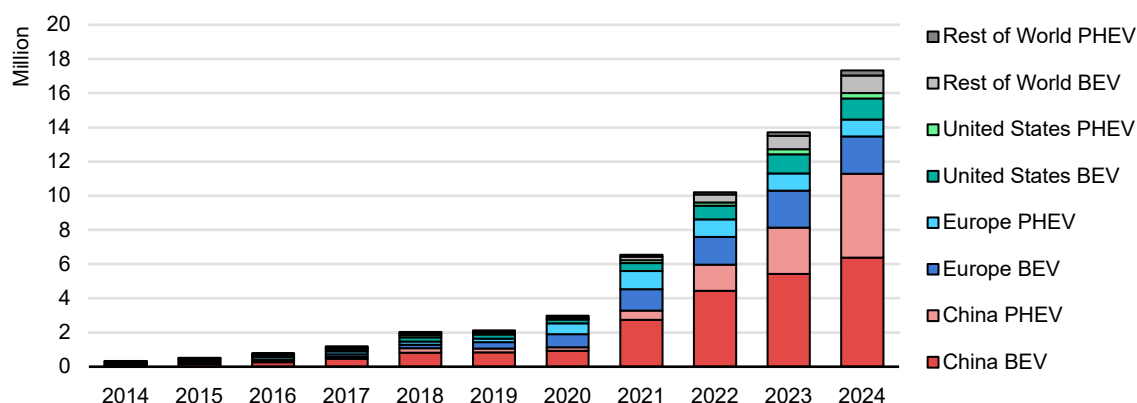
Electric car sales

Global electric car sales exceeded 17 million in 2024

More than 20% of new cars sold worldwide were electric

Electric car sales topped 17 million worldwide in 2024, rising by more than 25%.¹ Just the additional 3.5 million cars sold in 2024 compared to 2023 outnumber total electric car sales in the whole of 2020. China maintained its lead among major markets, with electric car sales exceeding 11 million – more than were sold worldwide just 2 years earlier. Global sales were slightly tempered by stagnating growth in Europe, as subsidies were phased out or reduced in several major markets, and as the EU CO₂ targets for cars remained the same between 2023 and 2024. Electric car sales continued to increase in the United States although growth was about one-quarter that of the previous year. Significantly, outside of these three major markets, there was a record increase in sales of nearly 40% to reach 1.3 million, closing in on the United States' sales of 1.6 million electric cars.

Global electric car sales, 2014-2024



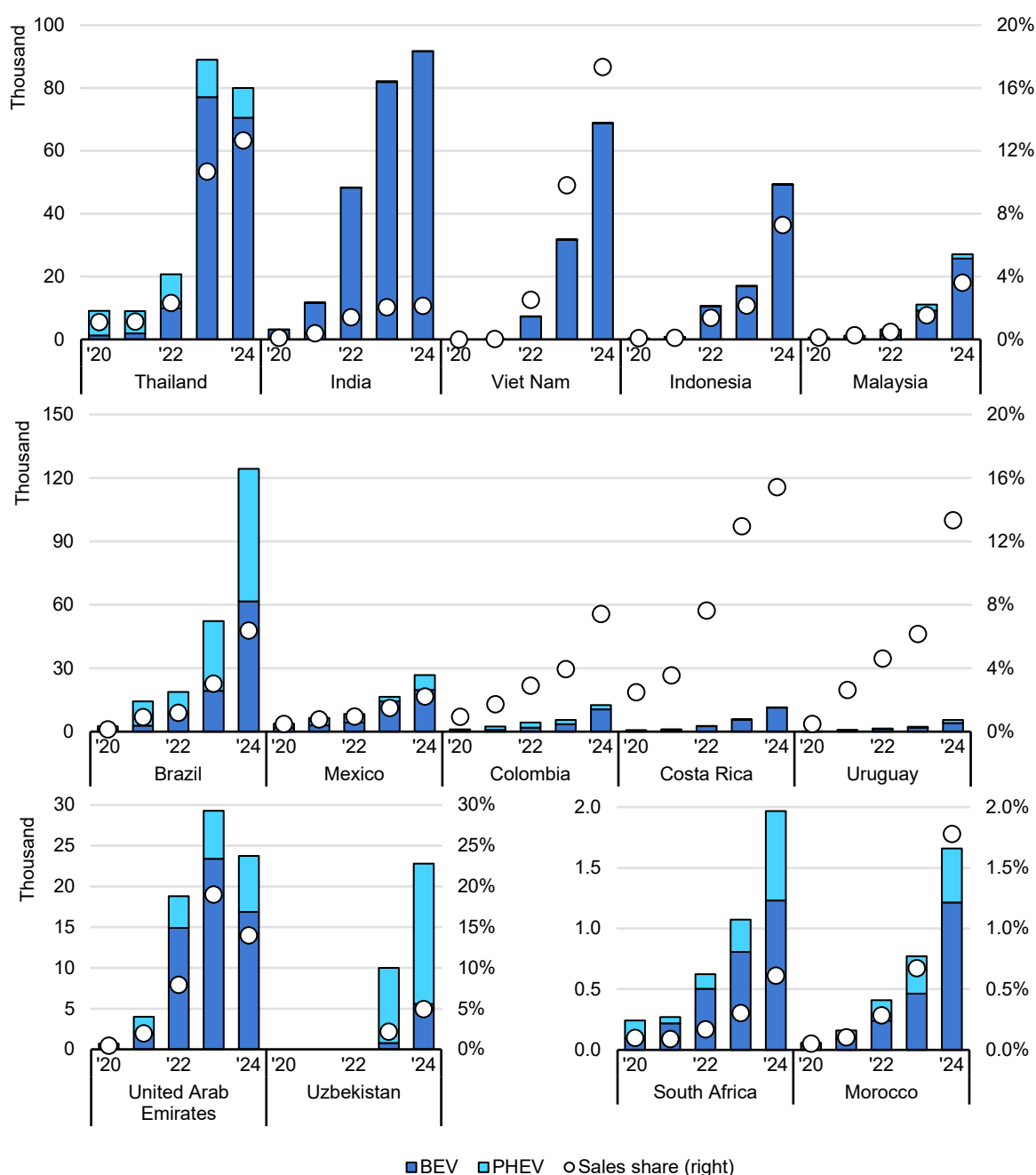
IEA. CC BY 4.0.

Notes: BEV = battery electric vehicle; PHEV = plug-in hybrid vehicle. Includes new passenger cars only.

Sources: IEA analysis based on country submissions and data from the European Automobile Manufacturers Association (ACEA), European Alternative Fuels Observatory (EAFO), EV Volumes and Marklines.

¹ In this report “sales” represents an estimate of the number of new vehicles hitting the roads. Where possible, data on new vehicle registrations is used. In some cases, only data on retail sales (such as sales from a dealership) are available. New car sales or registrations exclude used cars. Unless otherwise specified, the term electric vehicle is used to refer to both battery electric and plug-in hybrid electric vehicles but does not include fuel cell electric vehicles. For a brief description of the trends related to fuel cell electric vehicles, see the box at the end of [Section 4](#).

Electric car registrations and sales shares in selected countries, 2020-2024



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Notes: BEV = battery electric vehicle; PHEV = plug-in hybrid electric vehicle.

Sources: IEA analysis based on country submissions and data from ACEA, EAFO, EV Volumes, Marklines, [Asomove](#), [AleTech](#), [Andemos](#), OICA, [AFMA](#), [Gaikindo](#) and [AIVAM](#), sales of other macro regions can be found in the [Global EV Data Explorer](#).

Emerging electric car brands are seeing local and even international success

Recent years have seen the birth of several new electric car brands in emerging markets, such as VinFast in Viet Nam, Togg in Türkiye, and [Tito](#) in Argentina,

2. Trends in the electric car industry

Manufacturing and trade

Steady growth in global electric car production masks differences at the regional level

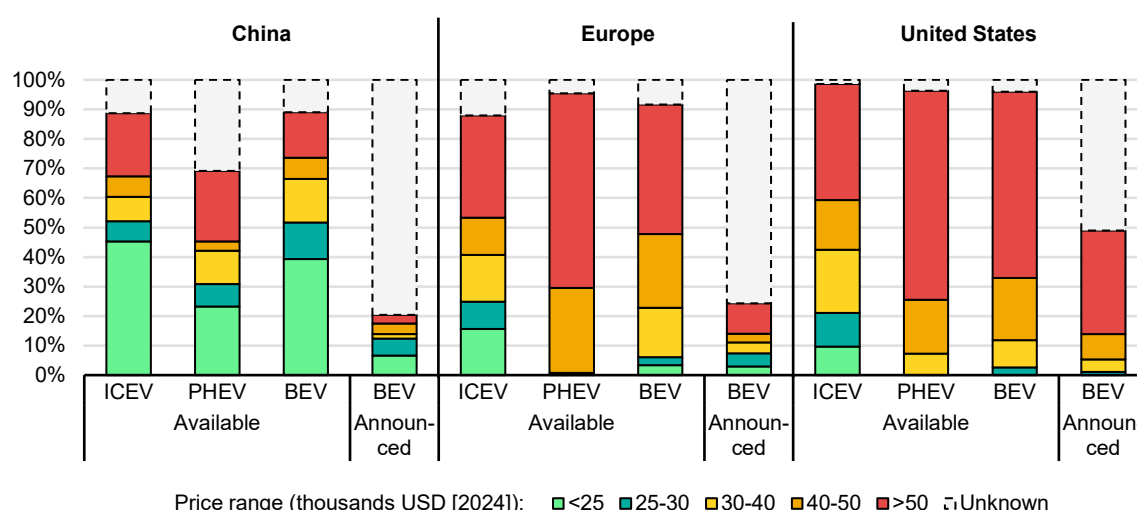
A total 17.3 million electric cars were produced worldwide in 2024, about one-quarter more than in 2023, largely as a result of increased production in **China**, which reached 12.4 million electric cars. China remains the world's electric car manufacturing hub, accounting for more than 70% of global production in 2024. Production in China has been increasingly shaped by the expansion of domestic manufacturers. In 2024, Chinese OEMs accounted for more than 80% of domestic production, up from roughly two-thirds in 2021. Despite the numerous recently announced foreign direct investment plans from Chinese OEMs, their overseas production has yet to ramp up. Electric car production by Chinese OEMs operating outside China accounted for less than 2% of their global output in 2024.

In the world's second-largest electric car manufacturing region, the **European Union**, production stagnated at 2.4 million cars in 2024, but surpassed domestic sales by more than 5%. Domestic carmakers were behind nearly 80% of the region's total output, but there were contrasting trends among EU OEMs. While German OEMs marked a 5% year-on-year increase in their EU output, other EU OEMs (Stellantis and Renault) saw their regional production drop by over 15%, producing about 420 000 electric cars, or less than 20% of the region's output. Meanwhile, there was a sixfold increase in EU production by US OEMs between 2021 and 2024, predominantly led by Tesla and Ford. This contributed to the share of foreign OEMs in EU production reaching about 20% in 2024.

Elsewhere in Europe, the United Kingdom saw its electric car output drop 30% year-on-year in 2024 to around 80 000 electric cars, while Türkiye's production grew to 45 000, with two-thirds produced by domestic manufacturer Togg.

However, the trend towards falling prices has been uneven across markets, due to differences in carmakers' pricing strategies, and in market maturity and the level of competition. For example, in China, the sales-weighted average (hereafter, "average") price of a battery electric SUV fell almost 10% year-on-year in 2024, partly due to the 30% decline in the battery pack price. Similarly, in the United States, a 15% decline in battery prices contributed to a 3% drop in the average purchase price of electric SUVs in 2024. In contrast, in Germany, the price of electric SUVs slightly increased in 2024, despite their battery pack prices declining 20%. This suggests that the battery pack price is not the only factor influencing EV prices: other component manufacturing costs, trim levels and carmakers' pricing strategies also play a significant role.

Price range distribution of available and announced car models in selected markets, 2024-2026



Price range (thousands USD [2024]): <25 25-30 30-40 40-50 >50 Unknown

IEA. CC BY 4.0.

Notes: ICEV = internal combustion engine vehicle; PHEV = plug-in hybrid electric vehicle; BEV = battery electric vehicle. "Available" includes models sold in 2024 in selected markets. Germany, the United Kingdom and Türkiye are used as proxy countries for available models in Europe. "Announced" only includes models with known release price and expected to be launched by the end of 2026.

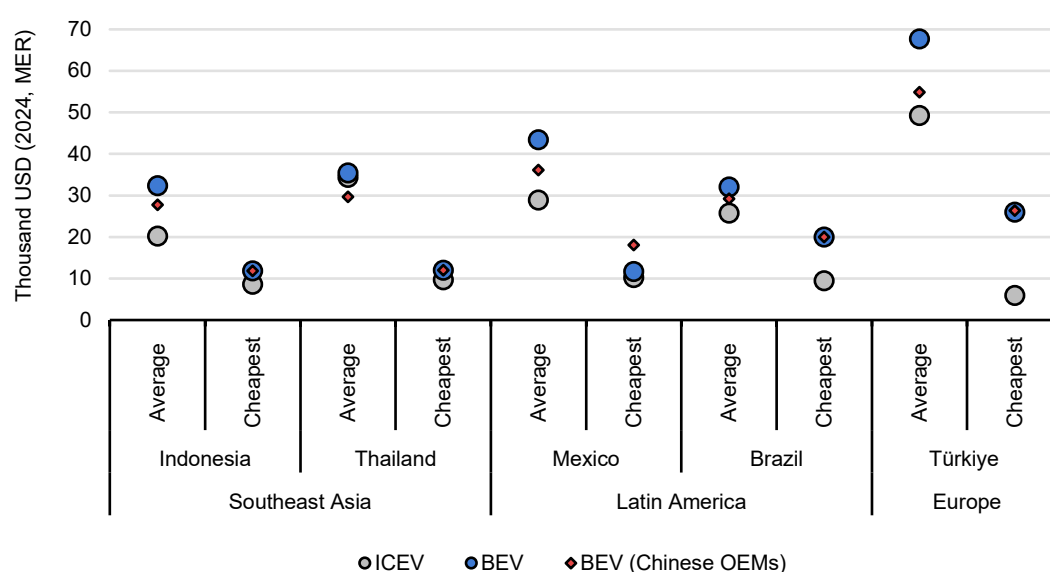
Source: IEA analysis based on data from S&P Global Mobility, EV Volumes and OEM announcements.

Availability of a wide range of affordable EV models will be key to unlocking mass-market adoption. In 2024, there were fewer BEV models available than ICEV models in the United States and Europe, and the range was skewed towards higher-end models with higher prices. Conversely, in China, the price distribution of available BEV models closely resembles that of ICEVs, with about 40% of available electric models priced below USD 25 000 (against 45% for ICEV models), and more than half below USD 30 000. In 2024, this price distribution was reflected in sales, with the median price paid for a battery electric car standing at around USD 24 000, about USD 700 less than for an ICE car. This trend

retailed at similar prices to the cheapest ICE models. Choosing an electric car in these markets could carry little to no price premium.

In all of the five emerging markets assessed – Brazil, India, Indonesia, Mexico and Thailand – electric models made by Chinese OEMs were, on average, cheaper than the average electric car, helping to drive uptake. The cheapest Chinese battery electric car was, in most cases, also the cheapest option on the market, so much that in **Thailand** in 2024, the average price of a Chinese EV was lower than the average price of a conventional car. In 2023 and 2024, more than four out of every five battery electric cars sold in Thailand were imported from China, bringing the average price of BEVs nearly in line with that of conventional cars. On average, medium and SUV model BEVs were more than 20% cheaper than their conventional equivalents, and the overall price premium of BEVs, across all car segments, stood below 5%.

Average and cheapest battery electric car price values by powertrain in selected emerging markets, 2024



IEA. CC BY 4.0.

Notes: ICEV = internal combustion engine vehicle; BEV = battery electric vehicle; OEM = original equipment manufacturer. Average price refers to the sales-weighted average price of cars per powertrain. The cheapest category shows the lowest price found in data.

Source: IEA analysis based on data from S&P Global Mobility.

In **Brazil**, where Chinese electric car imports increased to reach 85% of the country's EV sales in 2024, up from about 60% in 2023, the price gap between battery electric and ICE cars shrank from more than 100% to 25% over the same period. In 2024, not only was the cheapest battery electric car model produced in China by a Chinese OEM, but Chinese BEVs also retailed at less than half the price of those produced by non-Chinese OEMs on average (most of which belong

5. Outlook for electric mobility

Overview

In this part of the report, we focus on the outlook for electric mobility in road transport over the period to 2030. A scenario-based approach is used to explore the prospects for electric mobility, based on recent market trends, policy drivers and technology developments.

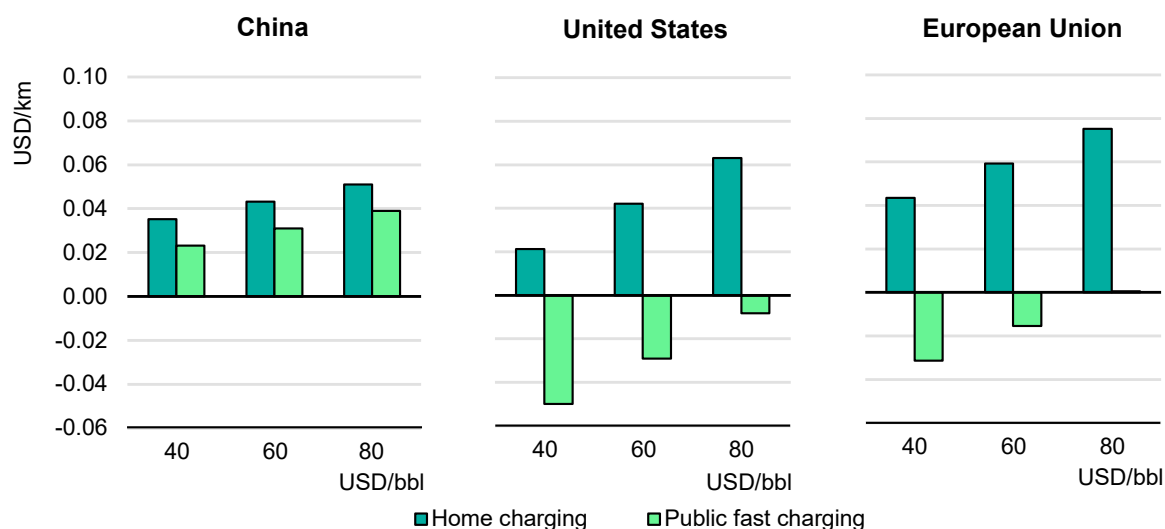
The purpose of scenario projections is to assess a plausible future for global electric vehicle (EV) markets and the potential implications. The scenario projections are not intended as predictions about the future. Rather, they aim to provide insights to inform decision-making by governments, companies and other stakeholders about the future of EVs.

In particular, the [Stated Policies Scenario](#) (STEPS) reflects the current policy landscape, taking into account existing policies and measures, as well as those that are under development. It includes current EV-related policies and announcements, regulations and investments, as well as market trends based on the expected impacts of technology developments, announced deployments and plans from industry stakeholders. The STEPS aims to hold up a mirror to the plans of policy makers and illustrate their consequences.

The projections for EV markets in this year's Global EV Outlook (GEVO-25) are limited to 2030 in the STEPS. As the IEA highlighted in its [2024 edition of Energy Technology Perspectives](#), energy, industrial and trade policies are increasingly interwoven and there are challenges and trade-offs between these key areas of public policy. The way these interactions play out for electric mobility over the longer term will therefore be assessed holistically in the context of the scenarios of the entire global energy system that the IEA will develop later in the year using its Global Energy and Climate Model (GEC-M). Key uncertainties for EV markets for the medium-term – such as those related to the evolution of trade and industrial policy, downside risks to the economic outlook and the impact of different levels of oil prices – are, however, presented in this chapter of the report.

The projections in the STEPS in GEVO-2025 consider historical market data and stated policies up until the end of February 2025. These scenario projections incorporate GDP assumptions from the International Monetary Fund and population assumptions from the United Nations, as described in the [2024 GEC-M documentation](#).

Electric car running cost savings compared to an internal combustion engine car for a range of illustrative oil prices, 2024

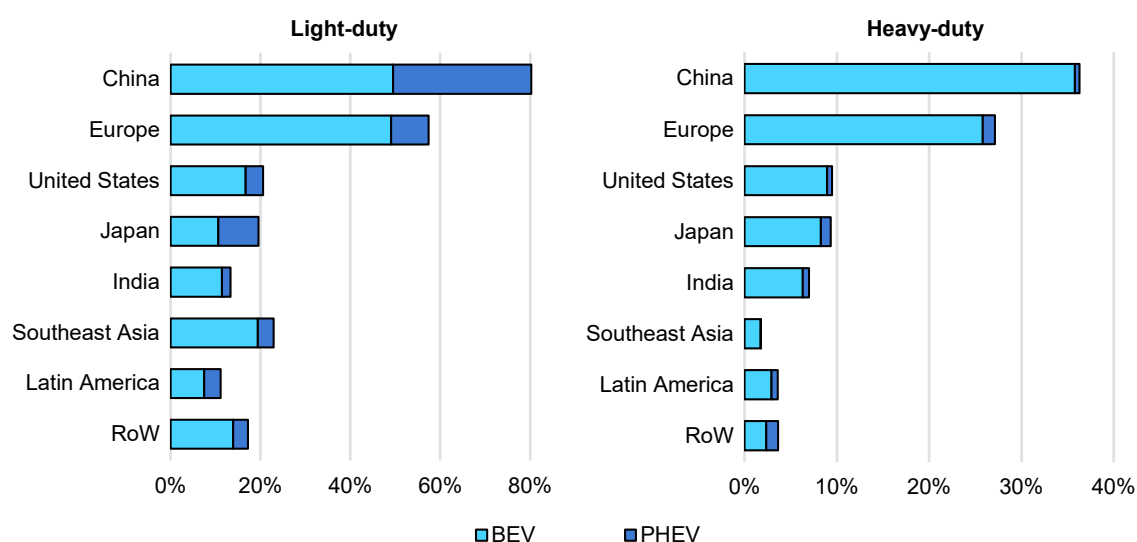


IEA. CC BY 4.0.

Notes: bbl = barrel. Negative savings indicate it costs more to run a battery electric car than an internal combustion engine car. Average on-road fuel consumption values in each region is assumed.

Vehicle outlook by region

Electric vehicle sales share by mode and region in the Stated Policies Scenario, 2030



IEA. CC BY 4.0.

Notes: BEV = battery electric vehicle; PHEV = plug-in hybrid electric vehicle; RoW = Rest of World. Light-duty refers to passenger light-duty vehicles and light commercial vehicles. Heavy-duty refers to buses, medium-duty and heavy-duty trucks. Regional projected EV sales and sales shares data can be explored in the interactive [Global EV Data Explorer](#).

7. Electric vehicle batteries

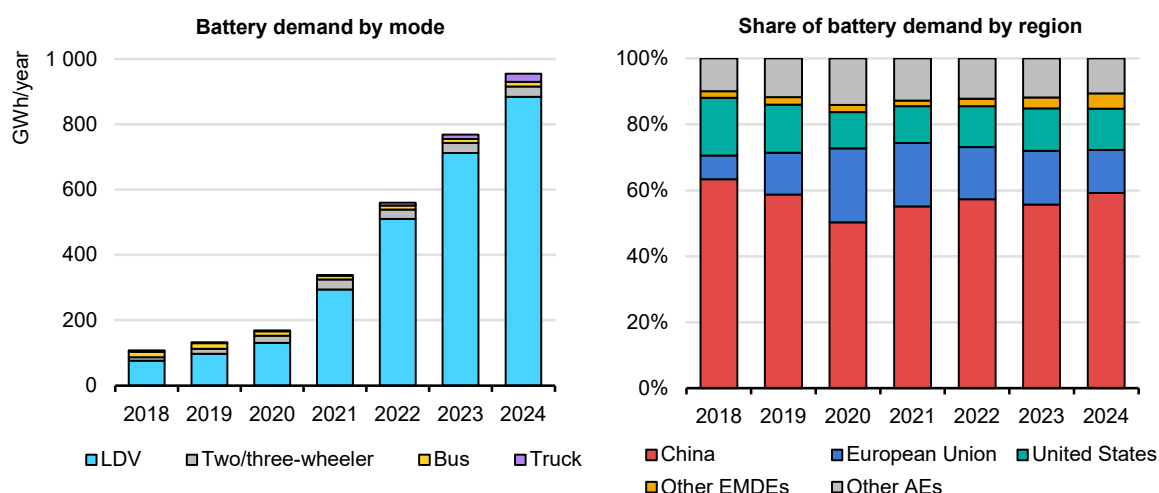
Trends in battery demand

Global battery demand for the energy sector hit the 1 TWh milestone in 2024

Electric cars remain the main driver of battery demand, but demand for trucks nearly doubled

Battery demand in the energy sector, for both EV batteries and storage applications, reached the historical milestone of 1 TWh in 2024. Demand for one average week alone in 2024 exceeded the total demand for an entire year just a decade earlier. Demand was largely driven by growth in EV sales, as demand for EV batteries grew to over 950 GWh – 25% more than in 2023. Electric cars remain the principal factor behind EV battery demand, accounting for over 85%. Compared to 2023, the sector whose demand grew the most was electric trucks, growing over 75% in 2024 to reach nearly 3% of global EV battery demand. Electric truck battery demand was driven by growth in China, but demand also ramped up in Europe (about 25%), which accounted for about 10% of the global total.

Electric vehicle battery demand by mode and region, 2018-2024



IEA. CC BY 4.0.

Notes: LDV = light-duty vehicle, including cars and vans; EMDEs = Emerging markets and developing economies; AEs = Advanced economies. Battery demand is defined here as the volume-weighted average battery size multiplied by vehicle sales by mode and region. It reflects the batteries installed in vehicles sold in each region, and not the battery demand for vehicles manufactured in each region. Electric vehicle and battery stockpiling are excluded from the analysis.

Source: IEA analysis based on [EV Volumes](#).

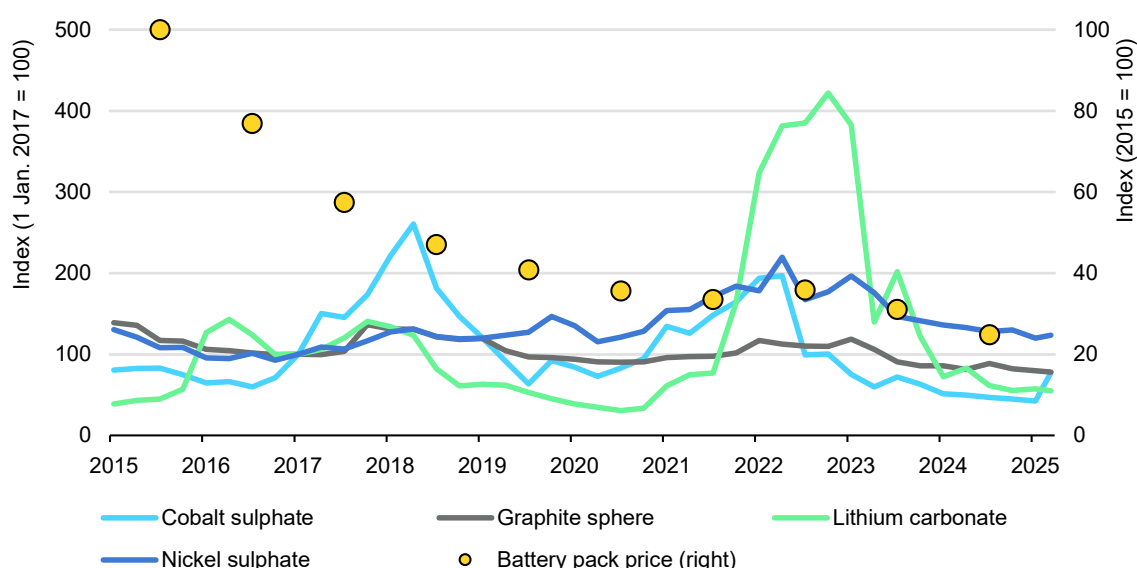
nearly 20% in 2024, reaching similar prices to those at the end of 2015, despite lithium demand in 2024 being about six times bigger than in 2015.

Low critical mineral prices are primarily driven by supply surplus, which is making it [difficult](#) for some mining companies to compete, thus increasing the level of supply chain concentration among established players. This surplus is [expected](#) to persist over the next few years, but low prices could discourage future investments and might cause supply [shortages](#) for [lithium](#) and [nickel](#) by 2030. In addition, the high [geographical](#) and [ownership](#) concentration of their supply chains may cause market distortions, increasing market risk.

An undersupply of lithium would push up prices, to the benefit of the mining sector but to the detriment of battery and EV makers, as well as final consumers. The recycling sector, which could help curb cost increases, would also benefit from higher mineral prices. However, due to feedstock limitations, it will take about a decade before recycling has a significant impact on reducing primary mineral demand (see the Box on [battery recycling](#) below).

[Technology innovation](#), particularly related to [sodium-ion batteries](#) or [direct lithium extraction](#), could be instrumental in reducing the risk of lithium undersupply and its potential impact, and avoiding price spikes similar to those seen in 2022. Additionally, vertical integration through [upstream investments](#) can help battery suppliers to lower production costs while safeguarding against the risk of volatile critical mineral prices.

Price of selected battery metals (left) and lithium-ion battery packs (right), 2015-2025



IEA. CC BY 4.0.

Notes: "Battery pack price" refers to the volume-weighted average pack price of lithium-ion batteries across the electric vehicles and battery storage sectors. 2025 refers to data up to the end of March 2025.

Sources: IEA analysis based on data from [Bloomberg](#) and [Bloomberg New Energy Finance](#) Lithium-Ion Price Survey (2024).

therefore help the European automotive industry to reduce costs and increase its competitiveness in the coming years. However, investments in LFP battery manufacturing in Europe might suffer as a result of the recent [proposition](#) from the Chinese government to limit exports of key battery technologies, including LFP cathode production and lithium processing. Overall, renewed interest in the LFP chemistry [exacerbates](#) the difficulties facing the European battery industry, which has historically focused on NMC batteries.

The year 2024 was a challenging one for the European battery industry, with both small and large producers suffering for different reasons. New players such as Northvolt, which filed for [bankruptcy](#) in the [United States](#) and [Sweden](#), faced serious difficulties in scaling up production, leading to insufficient manufacturing [yield](#) and high production costs. The case of Northvolt was further [aggravated](#) by limited experience, ambitious expansion plans, and excessively widened scope, leading to insufficient focus on scaling up high-quality battery cell production. Falling battery prices worldwide also weighed heavily on small producers, which have fewer resources and insufficient production volumes to withstand [lower profit margins](#), and are now turning to a more [cautious](#) growth path.

Korean companies, such as LG Energy Solution in Poland, are the largest battery producers in Europe, but their market share is also under pressure. Over the past 2 years, the market share of Korean manufacturers in the European Union has fallen from nearly 80% in 2022 to 60% in 2024, to the benefit of Chinese manufacturers. A key reason for this is the increased success of the LFP chemistry. In contrast, the share of Korean companies in the United States grew from about 20% in 2022 to over 35% in 2024, at the expense of Japanese manufacturers.

Korean manufacturers are rising to the challenge and are now [investing](#) in LFP, including through [innovation](#) efforts, and are scaling up LFP battery production, including in [Europe](#) and the [United States](#). LFP battery development is also advancing in Japan: In early 2025, Nissan [announced](#) plans to build an LFP battery plant after receiving [government certification](#) in late 2024. Yet Chinese manufacturers are continuing to innovate LFP [chemistry](#) and [performance](#) simultaneously, further raising the bar for other producers. At the same time, the Chinese government's [proposed](#) export restrictions on [advanced](#) LFP technologies could limit technology transfer.

In the United States, Panasonic – historically Tesla's main battery partner – remained the country's largest battery producer in 2024, accounting for nearly half of the electric car market. However, its share is declining as Korean producers like LG Energy Solution, Samsung and SKI invest heavily in the US market. Tesla also ramped up its own battery production in 2024, primarily for use in the Cybertruck.

Battery production and trade

The battery market is being reshaped by two distinct trends: increasing consolidation, and government-led efforts to geographically diversify battery supply chains. Whereas markets used to be regionalised and small, they are now global and very large, and uncertainty over technological development is giving way to standardisation. In this [new phase](#), economies of scale, partnerships along the supply chain, manufacturing efficiency, and the capacity to bring innovations swiftly to market will be even more crucial for manufacturers to remain competitive.

The battery supply chain remains geographically concentrated

The EV and battery supply chain becomes increasingly geographically concentrated when moving upstream from car manufacturing to battery cell and component production. China was responsible for 80% of global battery cell production in 2024, while the remainder was produced in the United States, the European Union, Korea and Japan. Importantly, the production of lithium-ion battery manufacturing [equipment](#) is also highly concentrated, with China, Korea, and Japan [leading](#) the market. China has also established a near monopoly on battery components production, supplying almost 85% of cathode active materials – including NMC and LFP chemistries – and over 90% of anode active material production, predominantly graphite. Outside of China, only Korea and Japan offer sizeable production capacity for cathode components. Korea also produces anodes, and Indonesia is [expected](#) to bring some diversification to the market. Nonetheless, China is set to remain the largest producer of batteries and their components by some distance in the medium term, based on announced projects and [competitive advantages](#).

The [geographical concentration](#) of [battery mineral](#) mining and refining needed for battery cathodes and anodes is also a concern. In 2023, Australia, Chile and China mined about 85% of global lithium, with almost 65% refined in China and another 25% in Chile. Indonesia accounted for over half of nickel mining in 2023, while China and Indonesia together refined more than 60%. In the same year, the Democratic Republic of Congo was home to almost two-thirds of the world's cobalt mining, though three-quarters of all cobalt refining was handled in China. Graphite supply, the only critical mineral used for anodes today, is even more concentrated, with China responsible for 80% of mining and over 90% of refining.

8. Outlook for energy demand

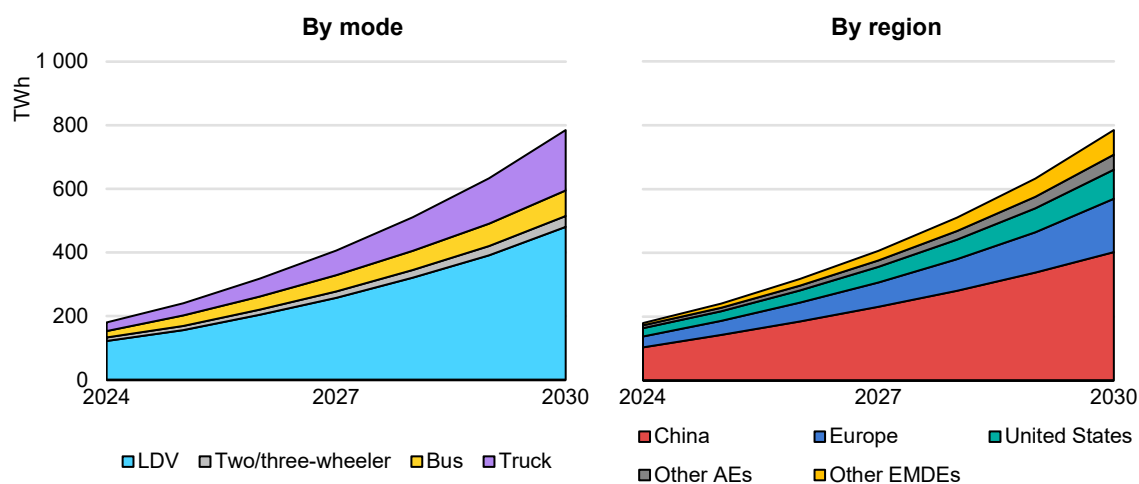
Electricity demand

Electric vehicles could account for more than 4% of European electricity demand by 2030

In 2024, the global fleet of EVs consumed around 180 TWh of electricity,¹ almost 60% more than the previous year. To put this in perspective, 180 TWh is more than the annual electricity consumption of Argentina. At the global level, EVs represented about 0.7% of final electricity consumption in 2024.

The stock of EVs is set to more than triple to 2030, but electricity demand could increase more than fourfold, reaching 780 TWh in the STEPS. This is driven by increasing consumption from electric trucks, as well as greater EV uptake in markets where people drive more per year. Total energy demand for road transport increases by only 5% in 2030 in the STEPS, while total road activity (in terms of vehicle kilometres travelled) increases by almost 20% during the same period, reflecting the greater energy efficiency of EVs.

Electricity demand by mode and by region in the Stated Policies Scenario, 2024-2030



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Notes: LDV = light-duty vehicle; AEs = advanced economies; EMDEs = emerging markets and developing economies. The analysis is carried out for each region in the transport model within the IEA's [Global Energy and Climate Model](#) (GEC-Model) separately and then aggregated for global results. Regional data can be interactively explored via the [Global EV Data Explorer](#).

¹ Electricity demand includes vehicle charging losses, which vary based on charging speed; it does not include transmission and distribution losses. For further information, see the [model documentation](#).

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